```
import warnings
warnings.filterwarnings("ignore")
import numpy as np
from sklearn.datasets import load_digits
from sklearn.model_selection import
train_test_split
from sklearn.preprocessing import
StandardScaler
from sklearn.neural_network import
MLPClassifier
from sklearn.linear_model import
LogisticRegression
from sklearn.svm import SVC
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score
digits = load_digits()
X = digits.data
y = digits.target
X_train, X_test, y_train, y_test = train_test_split(X,
y, test_size=0.2, random_state=42)
```

```
print("Training Data:")
print(X_train)
print("Testing Data:")
print(X_test)
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
nn = MLPClassifier(hidden_layer_sizes=(100,
100), max_iter=500, random_state=42)
nn.fit(X_train, y_train)
nn_pred = nn.predict(X_test)
nn_acc = accuracy_score(y_test, nn_pred)
lr = LogisticRegression(max_iter=500)
lr.fit(X_train, y_train)
lr_pred = lr.predict(X_test)
lr_acc = accuracy_score(y_test, lr_pred)
svm = SVC(kernel='linear', max_iter=5000)
svm.fit(X_train, y_train)
svm_pred = svm.predict(X_test)
svm_acc = accuracy_score(y_test, svm_pred)
dt = DecisionTreeClassifier(max_depth=10,
```

```
random_state=42)
dt.fit(X_train, y_train)
dt_pred = dt.predict(X_test)
dt_acc = accuracy_score(y_test, dt_pred)
print("MODEL ACCURACY COMPARISON\n")
print("Neural Network Accuracy :", nn_acc)
print("Logistic Regression Accuracy :", lr_acc)
print("SVM Accuracy :", svm_acc)
print("Decision Tree Accuracy :", dt_acc)
```

```
Training Data:
[[ 0.  0.  3. ... 13.  4.  0.]
 [ 0.  0.  9. ...  3.  0.  0.]
 [ 0.  0.  0. ...  6.  0.  0.]
 ...
 [ 0.  0.  9. ... 16.  2.  0.]
 [ 0.  0.  1. ...  0.  0.  0.]
 [ 0.  0.  1. ...  1.  0.  0.]]
Testing Data:
[[ 0.  0.  0. ... 14.  5.  0.]
 [ 0.  0. 11. ...  1.  0.  0.]
 [ 0.  0.  8. ...  8.  0.  0.]
```

```
 ...
 [ 0.  0.  7. ... 10.  0.  0.]
 [ 0.  0.  7. ... 16.  4.  0.]
 [ 0.  0. 14. ...  5.  0.  0.]]
```

MODEL ACCURACY COMPARISON

Neural Network Accuracy :

0.9777777777777777

Logistic Regression Accuracy :

0.9722222222222222

SVM Accuracy : 0.975

Decision Tree Accuracy : 0.85